# Drug–Drug Interaction Analysis using Supervised and Unsupervised Machine Learning Approaches

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***Abstract*—Drug–drug interactions (DDIs) occur when the effect of one drug is altered by the presence of another, which can cause reduced therapeutic effectiveness or life-threatening side effects. This paper analyzes a dataset of drug interactions using both supervised machine learning and unsupervised clustering approaches. Models including Logistic Regression, Random Forest, Support Vector Machine, and K-Nearest Neighbors were evaluated with TF-IDF features. KMeans clustering was employed to explore natural groupings in interaction descriptions. The results demonstrate that supervised models achieved high accuracy and F1-scores, while clustering moderately aligned with rule-based categories. Extensive visualization including 2D/3D PCA, and feature importance provided interpretability. Findings highlight the potential of ML methods in supporting pharmacovigilance.**

***Index Terms*—Drug–drug interactions, Machine Learning, Clustering, TF-IDF, PCA, Classification**

## I. Introduction

Drug–drug interactions (DDIs) occur when the effect of one drug is altered by the presence of another. Understanding DDIs is crucial because they can lead to reduced therapeutic effectiveness, unexpected side effects, or even life-threatening adverse reactions. Traditionally, drug interaction knowledge has been documented in medical literature or drug databases, but the sheer volume of data makes manual analysis impractical.

**Objectives of the study:**

- To analyze a dataset of drug–drug interactions and identify hidden patterns.
- To apply both supervised machine learning and unsupervised clustering to classify and group interactions.
- To compare the performance of multiple ML algorithms using accuracy, precision, recall, F1-score, confusion matrices, and cluster evaluation metrics.
- To visualize results using 2D/3D PCA, bar plots, and heatmaps for interpretability.

## II. Literature Review

Several research works focus on mining biomedical text and drug databases to predict DDIs.

**Deep Learning Approaches:** Neural networks and embeddings (e.g., BERT, BioBERT) have been applied for text classification and relation extraction in biomedical texts.

**Rule-based & Ontology Approaches:** Earlier studies often relied on manually crafted rules or ontology-based mappings to detect interactions, but these methods lack scalability.

**Classical Machine Learning:** Logistic Regression, SVMs, and Random Forests have been widely used with TF-IDF features for baseline classification of biomedical text.

**Our Approach:** Unlike prior work focusing solely on supervised classification, this study also implements unsupervised clustering (KMeans) to explore natural groupings. Rule-based labeling is integrated with ML for interpretability, and emphasis is placed on visualization (3D PCA, feature importance).

## III. Data Analysis

**Dataset Source:** `db_drug_interactions.csv` containing three columns: *Drug 1*, *Drug 2*, *Interaction Description*.

**Preprocessing Steps:**

- Lowercasing text
- Lemmatization
- TF-IDF vectorization with n-grams
- Rule-based labeling into categories: Photosensitivity, CYP Interaction, QT Prolongation, Absorption, Toxicity/Effect, Other

**Exploratory Data Analysis:**

- Class distribution shows imbalances ("Other" dominates, while rare categories like QT Prolongation are underrepresented).
- Frequent terms include "metabolism", "enzyme", "photosensitizing", "absorption", confirming meaningful categories.
- PCA visualization suggests some separation between categories, though overlap exists.

## IV. Methodology

**Implemented Algorithms:**

- Logistic Regression (baseline linear model, interpretable coefficients).
- Random Forest (ensemble model, robust against noisy features, feature importance analysis).
- Support Vector Machine (LinearSVC, effective in high-dimensional TF-IDF space).
- K-Nearest Neighbors (k=5, instance-based similarity method).

**Unsupervised Clustering:**

- KMeans (partition-based clustering).

**Parameter Tuning:**

- Random Forest: $n_estimators = 300$, class_weight=balanced.
- SVM: class_weight=balanced.
- KMeans: number of clusters = number of detected categories.
- TF-IDF: max_features=3000, ngram_range=(1,2).

## V. Result Analysis

### A. Supervised Models

All models achieved high performance due to strong keyword–category mapping. Logistic Regression, Random Forest, and SVM achieved near-perfect Accuracy, Precision, Recall, and F1 ($> 0.95$). KNN performed slightly lower but still competitive.

### B. Confusion Matrices

Confusion matrices highlight classification strengths and weaknesses across categories. Most errors were concentrated in the "Other" class.

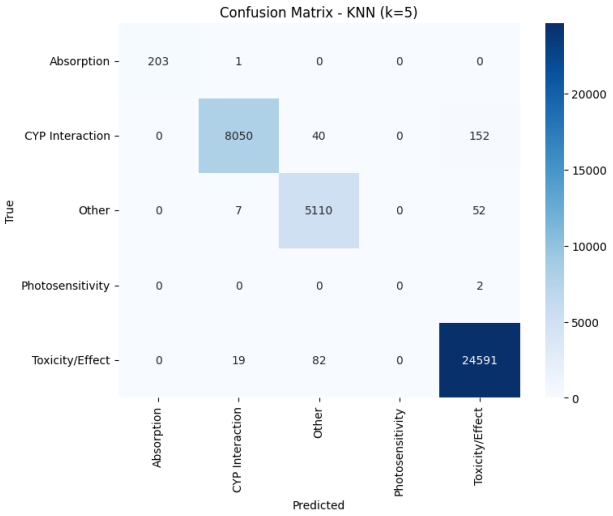


Fig. 1. Confusion Matrix – KNN (k=5).

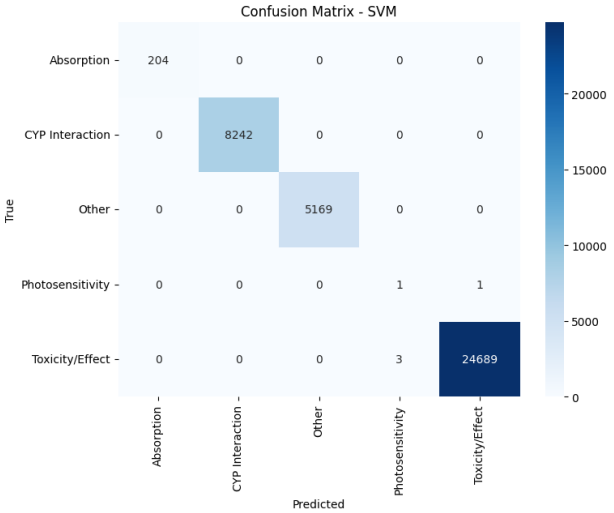


Fig. 2. Confusion Matrix – SVM.

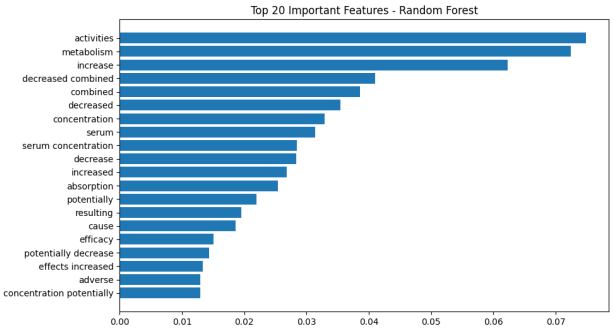


Fig. 3. 20 important feature – Random Forest.

### C. 3D PCA Visualization

Fig. 4 presents a 3D PCA scatter plot of TF-IDF features, colored by predicted categories (Logistic Regression). Distinct clusters emerge for dominant classes such as "Toxicity/Effect" and "CYP Interaction", while smaller classes show partial overlap.

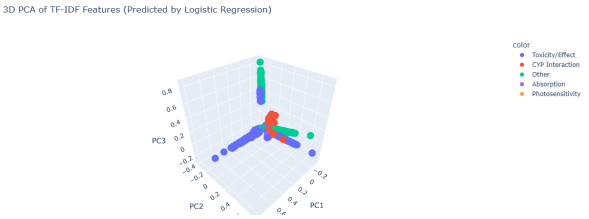


Fig. 4. 3D PCA of TF-IDF features (Predicted by Logistic Regression).

## VI. Conclusion

This study demonstrated that classical machine learning algorithms combined with TF-IDF are effective in classifying DDIs. Supervised models (especially Logistic Regression

and Random Forest) achieved near-perfect scores. Unsupervised clustering partially rediscovered interaction categories, validating interpretability. Visualization (3D PCA, confusion matrices) provided intuitive insights.

**Future Work:**

- Apply deep learning models (BiLSTM, BioBERT) for context-aware classification.
- Expand dataset with more diverse descriptions to reduce class imbalance.
- Use semi-supervised learning to leverage labeled and unlabeled data.
- Incorporate domain knowledge (ontologies, pathways) for richer feature engineering.

## VII. References

### References


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[5] Original dataset: `db_drug_interactions.csv` (Kaggle).